



Cycle Sequencing

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Helpful Hints or Important Considerations:

- Big Dye can be obtained from the reagent freezer in NHB W133. Please sign out any reagents you take using the iPad
- 10 minutes before starting, pull out Big Dye and primers (if necessary) to thaw.
- BigDye Buffer (BDB) is found in the fridge next to Wolverine. Do not freeze BDB.
- Once primers are thawed do not re-freeze. Primers can be stored @ 4°C for up to 2 weeks; subsequently throw away and thaw a new aliquot. Some people freeze primer aliquots after every usage. Follow your heart.

Materials & Reagents:

- Cleaned PCR product
- Big Dye v. 3.1 (Life Technologies)
- 5x Cycle Sequencing Buffer (Big Dye Buffer or BDB)
- Primer (10 µM)

Protocol:

1. Determine how many samples to sequence, including ~ 5% for pipetting loss. For example, to sequence all wells in a 96-well plate, you would need enough for 100 samples (96 plus 4 for pipetting loss).
2. In two 1.5 or 2 µl microcentrifuge tubes, mix all reagents. Use one tube for the Forward Reaction and one tube for the Reverse Reaction.

5X BigDye Buffer	1.75 µl per reaction
Primer (10 µM) F or R	0.25 µl per reaction
BigDye	0.5-1 µl per reaction
Nuclease-Free H ₂ O	6-6.5 µl per reaction (depending on BigDye volume used)

3. Pipette 9 µl of sequencing reagent cocktail into each well to be sequenced.
4. Pipette 1 µl cleaned PCR product into each well to be sequenced.
5. Spin plates.
6. Run plates on Thermocyclers using the profile:
 - 96°C 1 min
 - 30 cycles of: 96°C 30 sec
 - 50°C 30 sec
 - 60°C 2 min 30 sec
 - Hold @ 14°C indefinitely
7. If cleaning samples immediately, go to step 8, if not, plates can be stored at 4°C for up to 1 week before cleaning (although sequence quality may degrade over time). If you need to store samples longer, store at -20 C. However, for longer-term storage, it is better to clean sequencing products using the Sephadex protocol and store the dried samples in the dark at -20°C. Note that long-term



storage is not recommended because the fluorescent dyes will degrade over time, regardless of storage.

8. Clean samples using Sephadex protocol.